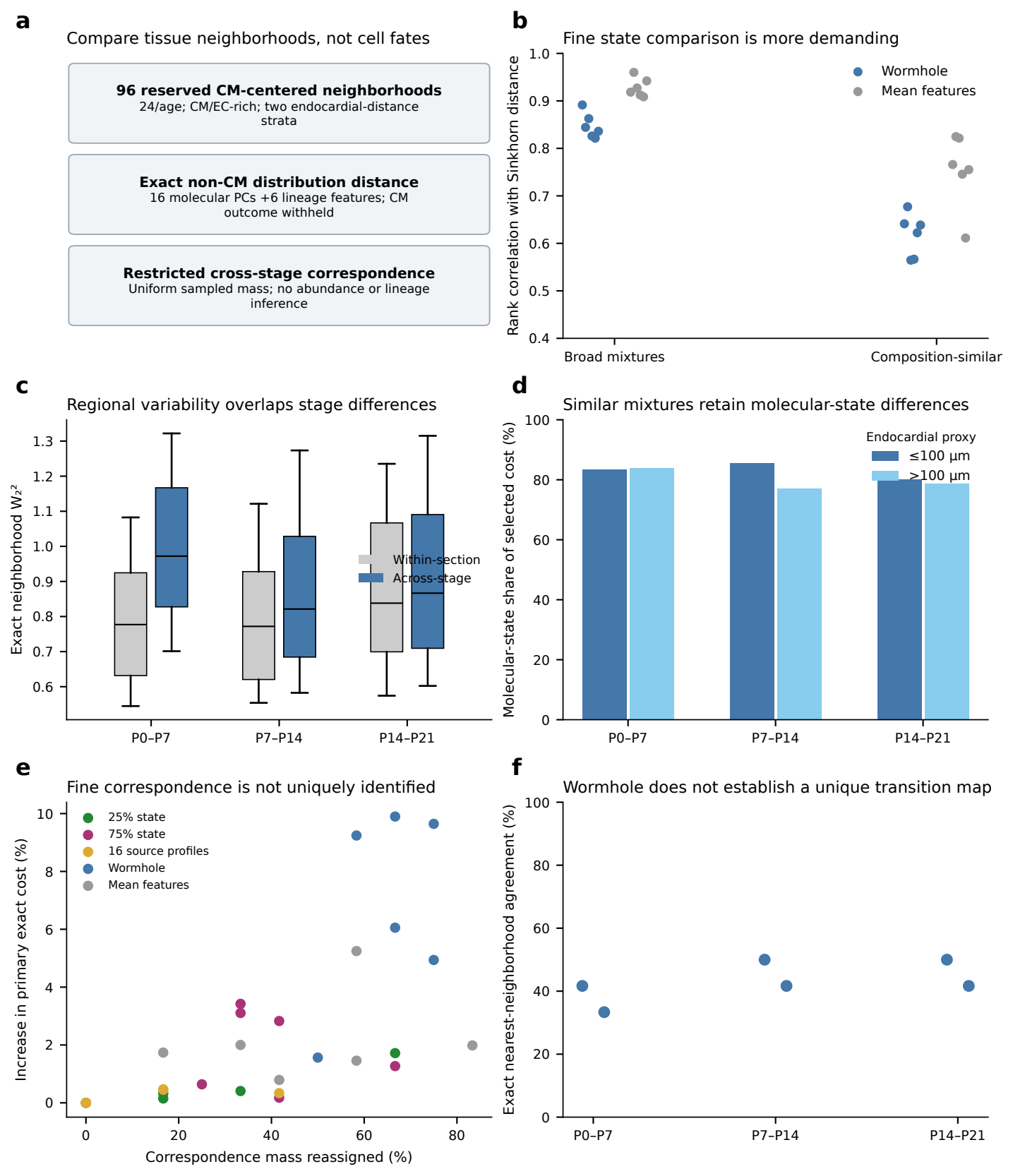
a
Compare tissue neighborhoods, not cell fates
96 reserved CM-centered neighborhoods
24/age; CM/EC-rich; two endocardial-distance strata
Exact non-CM distribution distance
16 molecular PCs +6 lineage features; CM outcome withheld
Restricted cross-stage correspondence
Uniform sampled mass; no abundance or lineage inference
b
Fine state comparison is more demanding
Rank correlation with Sinkhorn distance
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Wormhole
Mean features
Broad mixtures
Composition-similar
c
Regional variability overlaps stage differences
Exact neighborhood W_2^2
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
Within-section
Across-stage
P0–P7
P7–P14
P14–P21
d
Similar mixtures retain molecular-state differences
Molecular-state share of selected cost (%)
100
80
60
40
20
0
Endocardial proxy
≤100 µm
>100 µm
P0–P7
P7–P14
P14–P21
e
Fine correspondence is not uniquely identified
Increase in primary exact cost (%)
10
8
6
4
2
0
25% state
75% state
16 source profiles
Wormhole
Mean features
0
20
40
60
80
Correspondence mass reassigned (%)
f
Wormhole does not establish a unique transition map
Exact nearest-neighborhood agreement (%)
100
80
60
40
20
0
P0–P7
P7–P14
P14–P21